The Sen API

Interface Control Document

An example, generated from the data model by Sen. The reference that follows — every package, every type, every table — is written by the tool and is rewritten whenever the model changes.

This page is not. It is an ordinary Typst file included at a point the generated document leaves open, which is how a project puts its own cover, approvals and appendices around a reference it never has to edit.

Placeholder for your own project or product assets

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About this document

This is an example published with Sen's documentation. It was produced by running the model through `sen generate typst` and compiling the result, with no editing in between.

Everything from *Model overview* onwards is generated. It follows the model and only the model: if a type is added, renamed or given a new description, the next build says so, and nothing anyone wrote by hand is lost in the process.

This page, and the cover before it, are not generated. They are Typst files of your own that the document includes at points it leaves open for you:

`--front-matter`	A cover, before the contents.
`--before-reference`	Anything between the contents and the model — a scope, a set of conventions, an approvals table. This page.
`--after-reference`	Appendices, after the model.
`--style`	Every decision about how the document looks.

The generator does not read any of those files. It writes an `#raw("#include")` and Typst resolves it when the document is compiled, so a cover page can be anything Typst can set — and changing it never means regenerating the model.

Model overview

343 types in 14 packages. 14 classes in 12 independent hierarchies.

Packages

Package	Types	Object classes	Fixed records	Enumerations	Variant records	Arrays	Optionals
components	—	—	—	—	—	—	—
ether	31	—	20	1	3	4	3
explorer	1	—	1	—	—	—	—
influx	3	—	1	1	—	1	—
jsonrpc	41	2	17	—	1	8	13
logmaster	3	2	1	—	—	—	—
py	3	1	1	—	—	1	—
recorder	6	1	2	1	—	2	—
replayer	4	2	1	1	—	—	—
rest	37	—	20	5	—	9	3
shell	27	1	20	3	1	2	—
term	8	1	2	2	—	3	—
db	3	—	3	—	—	—	—
kernel	160	4	80	23	8	37	8
log	16	—	11	1	1	3	—

Class hierarchy across all packages

components.jsonrpc.JsonRpcServer	15
components.jsonrpc.StaticFileServer	18
components.logmaster.LogMaster	26
components.logmaster.Logger	26
components.py.PythonInterpreter	27
components.recorder.Recorder	28
components.replayer.Replay	30
components.replayer.Replayer	30
components.shell.Shell	38
components.term.Term	43
kernel.KernelApi	46
kernel.VirtualClock	46
├── kernel.VirtualKernelClock	46
└── kernel.VirtualMasterClock	46

Reading the tables

Every property carries three flags, in this order.

Flag	Meaning
`RO`	Read-only: the value is published by whoever owns it and cannot be set from outside.
`RW`	Read-write: the value can be set as well as read.
`D`	Dynamic: the value changes over the life of an instance.
`S`	Static: the value is fixed once an instance exists.
`C`	Confirmed: delivery is acknowledged, and the update is retried until it arrives.
`BE`	Best effort: the update is sent once and may be lost.

components

ether

31 types.

Fixed records

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ByteRange	Range of values of a u8	8
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DiscoveryHubAddress	TCP-based discovery hub parameters	8
Endpoint		8
Ephemeral	Let the operating system select an ephemeral port	8
Hello		9
MulticastAddressRange	Inclusive IPv4 multicast address range	9
MulticastDiscovery	Multicast-based discovery parameters	9
PinnedPort	Bind exactly this port, fail closed if taken	9
PortConfig	This configuration is per process and does not need to match across Sen instances. Exclusions apply to a pinned port, not to an ephemeral one: an ephemeral bind asks the OS for any free port and does not learn which it was given. udpUnicast cannot be pinned. Its socket is per peer and UDP has no four-tuple to demultiplex on, so a second peer either fails to bind or silently takes the first peer's traffic.	9
PortRange	Inclusive port range	9
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Ready		10
SessionPresenceBeam		10
TcpDiscovery	TCP-based discovery parameters	10

BusConfig STRUCTURE

Remember to set these to the same value in all related Sen instances

Name	Type	Description
`multicastRange`	MulticastRange	Defaults to the whole of 239.0.0.0/8: [239], [0-255], [0-255], [0-255]. RFC 2365 assigns only 239.192.0.0/14, which holds 262144 addresses and is not enough to keep many bus names apart. Set [239], [192-195], [0-255], [0-255] to stay inside it. Remember to keep a range wide enough for avoiding collisions if you have many buses.
`multicastExclusions`	MulticastAddressExclusions	Address ranges that must not be allocated
`multicastPort`	u16	Defaults to 50985 Change this in case you have limited control over the infrastructure.
`multicastDisabled`	bool	If true, no multicast traffic is performed and all the bus settings above become meaningless. Defaults to false.

Named by Configuration.

BusJoined STRUCTURE

Name	Type	Description
`participantId`	u32	
`busId`	u32	
`busName`	string	

Named by ControlMessage.

`BusLeft` STRUCTURE

Name	Type	Description
`participantId`	u32	
`busId`	u32	
`busName`	string	

Named by ControlMessage.

`ByteRange` STRUCTURE

Range of values of a u8

Name	Type	Description
`min`	u8	minimum range
`max`	u8	needs to be >= min

Named by MulticastRange.

`Configuration` STRUCTURE

Overall component configuration

Name	Type	Description
`discovery`	DiscoveryConfig	Discovery-related parameters
`busOutQueue`	QueueConfig	Queue for broadcasting to buses
`processUdpOutQueue`	QueueConfig	Queue for communicating with processes over UDP
`processTcpOutQueue`	QueueConfig	Queue for communicating with processes over TCP
`busConfig`	BusConfig	Bus transport configuration
`portExclusions`	PortExclusions	Port ranges that this process must not use
`portConfig`	PortConfig	If not present, all port bindings are Ephemeral
`runDiscoveryHub`	u16	If present, run the TCP discovery hub on this port
`networkDevice`	string	If present, the network device to use

`DiscoveryHubAddress` STRUCTURE

TCP-based discovery hub parameters

Name	Type	Description
`port`	u16	Port where the discovery hub is running
`host`	string	name or IP of the host where the discovery hub is running

Named by TcpDiscovery.

`Endpoint` STRUCTURE

Name	Type	Description
`ip`	u32	
`port`	u16	

Named by EnpointList.

`Ephemeral` STRUCTURE

Let the operating system select an ephemeral port

Named by PortBinding.

`Hello` STRUCTURE

Name	Type	Description
`info`	sen.kernel.ProcessInfo	
`udpPort`	u16	
`version`	ProtocolVersion	

Named by ControlMessage.

`MulticastAddressRange` STRUCTURE

Inclusive IPv4 multicast address range

Name	Type	Description
`min`	string	
`max`	string	

Named by MulticastAddressExclusions.

`MulticastDiscovery` STRUCTURE

Multicast-based discovery parameters

Name	Type	Description
`beamPeriod`	sen.Duration	How often are we beaming our presence
`beamExpirationTime`	sen.Duration	Expiry time used by the beam tracker
`mcastGroup`	string	Defaults to '239.255.0.44'
`port`	u16	Defaults to 60543
`device`	string	Device to use for the network communication
`allowVirtualInterfaces`	bool	If true, virtual (NO_CARRIER) interfaces are also considered

Named by DiscoveryConfig.

`PinnedPort` STRUCTURE

Bind exactly this port, fail closed if taken

Name	Type	Description
`port`	u16	

Named by PortBinding.

`PortConfig` STRUCTURE

This configuration is per process and does not need to match across Sen instances. Exclusions apply to a pinned port, not to an ephemeral one: an ephemeral bind asks the OS for any free port and does not learn which it was given. udpUnicast cannot be pinned. Its socket is per peer and UDP has no four-tuple to demultiplex on, so a second peer either fails to bind or silently takes the first peer's traffic.

Name	Type	Description
`tcpAcceptor`	PortBinding	inbound TCP listener
`udpUnicast`	PortBinding	per-peer UDP unicast; must be Ephemeral
`tcpSource`	PortBinding	outbound TCP source port

Named by Configuration.

`PortRange` STRUCTURE

Inclusive port range

Name	Type	Description
`min`	u16	
`max`	u16	

Named by PortExclusions.

`ProbePortRange` STRUCTURE

Bind within [min,max], random start, walk forward

Name	Type	Description
`min`	u16	
`max`	u16	

Named by PortBinding.

`ProtocolVersion` STRUCTURE

Name	Type	Description
`kernel`	u32	
`ether`	u32	

Named by Hello.

`QueueConfig` STRUCTURE

Name	Type	Description
`evictionPolicy`	QueueEvictionPolicy	What to do when a queue is full
`maxSize`	u64	Maximum element count, 0 means unbounded
`warningLevel`	u64	Element count where warning will be issued, 0 means none

Named by Configuration.

`Ready` STRUCTURE

Named by ControlMessage.

`SessionPresenceBeam` STRUCTURE

Name	Type	Description
`protocolVersion`	u16	
`info`	sen.kernel.ProcessInfo	
`beamPeriod`	sen.Duration	
`endpoints`	EnpointList	

`TcpDiscovery` STRUCTURE

TCP-based discovery parameters

Name	Type	Description
`beamPeriod`	sen.Duration	How often are we beaming our presence
`beamExpirationTime`	sen.Duration	Expiry time used by the beam tracker
`hubAddress`	DiscoveryHubAddress	Where to find the discovery hub
`allowVirtualInterfaces`	bool	If true, virtual (NO_CARRIER) interfaces are also considered

Named by DiscoveryConfig.

Enumerations

QueueEvictionPolicy	What to do when a queue is full	11

`QueueEvictionPolicy` ENUMERATION

What to do when a queue is full

Held as `u32`.

dropOldest	0
dropNewest	1

Named by QueueConfig.

Variant records

ControlMessage		11
DiscoveryConfig	Configuration parameters related to the discovery protocol	11
PortBinding		11

`ControlMessage` VARIANT

Type	Description
Hello	
Ready	
BusJoined	
BusLeft	

`DiscoveryConfig` VARIANT

Configuration parameters related to the discovery protocol

Type	Description
MulticastDiscovery	
TcpDiscovery	

Named by Configuration.

`PortBinding` VARIANT

Type	Description
Ephemeral	
PinnedPort	
ProbePortRange	

Named by PortConfig.

Arrays

EnpointList		11
MulticastAddressExclusions		12
MulticastRange	Range of values of a IPv4 multicast address	12
PortExclusions		12

`EnpointList` SEQUENCE

```
sequence<Endpoint, 27> EnpointList;
```

Named by SessionPresenceBeam.

MulticastAddressExclusions SEQUENCE

```
sequence<MulticastAddressRange> MulticastAddressExclusions;
```

Named by BusConfig.

MulticastRange SEQUENCE

Range of values of a IPv4 multicast address

```
array<ByteRange, 4> MulticastRange;
```

Named by BusConfig.

PortExclusions SEQUENCE

```
sequence<PortRange> PortExclusions;
```

Named by Configuration.

Optionals

MaybeDeviceName 12
MaybeDiscoveryHubPort 12
MaybePortConfig 12

MaybeDeviceName OPTIONAL

```
optional<string> MaybeDeviceName;
```

MaybeDiscoveryHubPort OPTIONAL

```
optional<u16> MaybeDiscoveryHubPort;
```

MaybePortConfig OPTIONAL

```
optional<PortConfig> MaybePortConfig;
```

explorer

1 types.

Fixed records

Configuration Overall component configuration 13

`Configuration` STRUCTURE

Overall component configuration

Name	Type	Description
`layoutFile`	string	the imgui layout file. Defaults to ""

influx

3 types.

Fixed records

Configuration Overall component configuration 14

Configuration STRUCTURE

Overall component configuration

Name	Type	Description
`samplingPeriod`	sen.Duration	sampling period for the recordings
`selections`	SelectionList	objects to be monitored (using sql)
`telegrafAddress`	string	ip or name of the Telegraf server
`telegrafPort`	u16	port used by the Telegraf server
`protocol`	Protocol	protocol used
`batchSize`	u32	points to batch before emission (0 means no batching)

Enumerations

Protocol 14

Protocol ENUMERATION

Held as u8.

udp	0
tcp	1

Named by Configuration.

Arrays

SelectionList 14

SelectionList SEQUENCE

```
sequence<string> SelectionList;
```

Named by Configuration.

jsonrpc

41 types.

Object classes

JsonRpcServer	JsonRpcServer -- the JSON-RPC over WebSocket external API, expressed as a Sen class. One instance per connected client lives on `local.jsonrpc`. Methods are the RPC surface; events are the server-pushed notifications. Wire framing (envelope, id, error mapping) is the dispatcher's job, not this class's. Wire-shape convention: JSON-RPC named params (`"params": { ... }`) use the same field names as the STL method's argument list. The dispatcher treats every method's params object as `{ <argName>: <value>, ... }` and walks `method->getArgs()` to build the VarList in declaration order. That lets the inbound dispatch path go through Sen's meta machinery with no per-method translation.	15
StaticFileServer	Server-wide registry of in-memory static-file bundles, served alongside the WebSocket+JSON-RPC surface by the same uWS app. Internal-only: lives on the configurable control bus (defaults to `local.jsonrpc_control`), not on the data-plane bus that hosts authenticated per-connection JsonRpcServer instances. The component publishes exactly one instance.	18

JsonRpcServer CLASS

JsonRpcServer -- the JSON-RPC over WebSocket external API, expressed as a Sen class. One instance per connected client lives on `local.jsonrpc`. Methods are the RPC surface; events are the server-pushed notifications. Wire framing (envelope, id, error mapping) is the dispatcher's job, not this class's. Wire-shape convention: JSON-RPC named params (`"params": { ... }`) use the same field names as the STL method's argument list. The dispatcher treats every method's params object as `{ <argName>: <value>, ... }` and walks `method->getArgs()` to build the VarList in declaration order. That lets the inbound dispatch path go through Sen's meta machinery with no per-method translation.

Name	Type	Flags	Description
`identity`	Identity	RO S BE	Identity of the connected client.

Method	Signature	Description
`ping`	`() → string`	Liveness probe.
`listTopology`	`() → SessionInfoList`	Snapshot of the currently-detected sessions, each with the buses currently detected inside it. The kernel's discovery surface is asynchronous; this returns whatever was visible at the moment of the call. Use `subscribeTopology` for live updates.
`subscribeTopology`	`()`	Sticky subscription to topology changes. Idempotent per connection. Wires a `topologyChanged` push immediately with the current full snapshot, then again on every session or bus add/remove. Cancel with `unsubscribeTopology`; the subscription is also torn down automatically when the connection closes.
`unsubscribeTopology`	`()`	Cancel the topology subscription created by `subscribeTopology`. Idempotent.
`getTypes`	`() → sen.kernel.StringList`	Returns the qualified names of every custom type registered with the kernel. Catalog only -- pair with `getType` to fetch a chosen spec. The bundled-with-objects path on `interestUpdate.types` remains the primary way clients learn about types they observe.

Method	Signature	Description
getType	(qualifiedName: string, withSchema: bool) → TypeLookupResult	Look up one type by qualified name. Returns the CustomTypeSpec; when `withSchema` is true, also returns the JSON-Schema fragment for the type (empty string otherwise). Dependency chasing is the client's job: each schema fragment carries `$ref` strings keyed by qualified name, resolved against whatever schema cache the client maintains.
createInterest	(interestName: string, query: string, subscribe: SubscribeBlock, withSchemas: bool)	Register a sticky interest defined by an STL query. Optional `subscribe` block pre-subscribes to properties/events on every matched object, including ones that arrive later. When `withSchemas` is true, every subsequent `interestUpdate` for this interest carries the JSON-Schema fragment for each previously-unseen type alongside the existing `CustomTypeSpec`. Re-registering the same name = invalidParams. Wire-ordering contract: the initial `interestUpdate.added` for the current match set is emitted *before* this RPC's response lands on the wire. Clients must accept pushes for an interest whose response has not yet been observed.
releaseInterest	(interestName: string)	Drop an interest and every subscription living under it.
listObjects	(interestName: string) → ObjectInfos	List the names and classes of objects currently in an interest's match set.
getProperty	(interestName: string, objectName: string, propertyName: string) → string	Read a property's current value, synchronously.
getObjectsBatchState	(interestName: string, objectNames: sen.kernel.StringList, propertyNames: sen.kernel.StringList) → ObjectStateList	Batched per-object snapshot. Walks the interest's match set in one pass instead of forcing the caller to issue N*M `getProperty` round-trips. Omitting `objectNames` reads every object currently matched; omitting `propertyNames` reads every property of each object (the class hierarchy is walked, so inherited members are included). Names from `objectNames` that aren't currently in the match set are silently skipped -- the caller compares its requested list against the returned `objectName`s to detect omissions. Names from `propertyNames` that don't exist on a given object's class are recorded per-object under `errors`, so a heterogeneous batch still returns useful partial results. Bounded: the call rejects with `invalidParams` when the resolved object set exceeds 512 entries (either via an explicit

Method	Signature	Description
		`objectNames` list of that length, or via an implicit walk over a match set that has grown that large). Page the request by passing `objectNames`.
`setProperty`	`(interestName: string, objectName: string, propertyName: string, value: string)`	Write a property's value (sugar over invoking the generated setter). Hand-routed in `Dispatcher::handleSetProperty`: the dispatcher issues `invokeUntyped` against the property's setter and wires the response through the kernel callback in a single tick. The typed `setPropertyImpl` override in the C++ `Server` exists only to satisfy the codegen-generated abstract base and is unreachable at runtime. See `architecture.md` "Method dispatch" for why this method (and `invoke`) bypass the generic meta path.
`subscribeProperty`	`(interestName: string, objectName: string, propertyName: string, maxRateHz: f64)`	Subscribe to propertyChanged notifications for one (interest, object, property). Sticky: persists across object churn; auto-rewires when the named object (re)enters the match set. Snapshot of the current value is delivered immediately after wiring. The throttle is per-object, shared across every property/event subscribed for the same object. An explicit `maxRateHz` overrides the current rate for the object; omitting it preserves whatever rate is already in effect (so adding a per-property sub doesn't accidentally retune existing subs back to unlimited).
`unsubscribeProperty`	`(interestName: string, objectName: string, propertyName: string)`	Cancel a property subscription previously created with subscribeProperty.
`subscribeEvent`	`(interestName: string, objectName: string, eventName: string)`	Subscribe to eventTriggered notifications for one (interest, object, event).
`unsubscribeEvent`	`(interestName: string, objectName: string, eventName: string)`	Cancel an event subscription previously created with subscribeEvent.
`subscribeAll`	`(interestName: string, objectName: string, maxRateHz: f64)`	Subscribe to every property and every event of one named object. Wildcard expansion against the object's actual class (including parent classes). Object must currently be in the match set; absent objects are invalidParams.
`unsubscribeAll`	`(interestName: string, objectName: string)`	Cancel an all-members subscription previously created with subscribeAll.
`invoke`	`(interestName: string, objectName: string, methodName: string, argsJson: string) → string`	Call a method on a kernel object and await its return value. Hand-routed in `Dispatcher::handleInvoke`: validates the args, issues `invokeUntyped`, and wires the response through the kernel callback in a single tick. The typed

Method	Signature	Description
		`invokeImpl` override in the C++ `Server` exists only to satisfy the codegen-generated abstract base and is unreachable at runtime. See `architecture.md` "Method dispatch".

Event	Payload	Description
`propertyChanged`	`(interestName: string, objectName: string, values: PropertyValueList, timestamp: sen.TimeStamp)`	Bundled per-object snapshot of changed properties. Per-object bundling, not per-property: any property change on one object lands in a single envelope.
`eventTriggered`	`(interestName: string, objectName: string, eventName: string, args: string, timestamp: sen.TimeStamp)`	An event fired on one object in one interest. The event signature is decided by the source object's class, not by JSON-RPC.
`interestUpdate`	`(interestName: string, added: AddedObjectList, removed: sen.kernel.StringList, types: sen.kernel.CustomTypeSpecList, typeSchemas: string)`	Match-set delta for one interest. The dispatcher tracks knownTypes / knownSchemas per connection and only re-sends specs / schemas new to the client.
`topologyChanged`	`(sessions: SessionInfoList)`	Full snapshot of the topology after a session or bus add/remove. Pushed once on `subscribeTopology` (with the current state) and again on every change. Snapshot form rather than deltas: topology changes are rare, and the snapshot keeps client logic trivial and idempotent.
`notificationsDropped`	`(count: u64)`	Sent once when outbound backpressure clears on this connection, reporting how many unreliable notifications were dropped while the buffer was above the high watermark. Emitted reliably on purpose: the notice about dropped messages must not be droppable. A client that receives this has an incomplete picture and cannot tell which updates it missed -- only that it missed some. The correct response is to re-read what it displays (getObjectsBatchState over the affected interests) rather than to wait for the next change, which may never come for a property that has stopped moving.

`StaticFileServer` CLASS

Server-wide registry of in-memory static-file bundles, served alongside the WebSocket+JSON-RPC surface by the same uWS app. Internal-only: lives on the configurable control bus (defaults to `local.jsonrpc_control`), not on the data-plane bus that hosts authenticated per-connection JsonRpcServer instances. The component publishes exactly one instance.

Name	Type	Flags	Description
`registeredBundles`	RegisteredBundleInfoList	RO D C	Currently-registered bundles. Externally read-only; the component mutates this whenever a bundle is registered or unregistered. Subscribe to receive notifications on every change.

Method	Signature	Description
`registerStaticBundle`	`(bundle: StaticBundle) → u64`	Register a bundle. The bundle is copied into the registry; the caller may release the argument

Method	Signature	Description
		afterwards. Routes are wired up asynchronously on the uWS thread; the returned bundleId is stable and may be used immediately with `unregisterStaticBundle`.
`unregisterStaticBundle`	`(bundleId: u64)`	Stop serving the bundle. The slot in the registry is nulled; the underlying uWS route remains registered (uWS does not expose a clean route-removal API) but becomes inert and returns 404. No-op if `bundleId` is unknown.

Fixed records

AddedObjectEntry	One object newly entering an interest's match set.	19
Configuration	JSON-RPC over WebSocket component configuration. Only `address` and `port` are required; every other field falls back to a documented default.	20
ConnectionLimits	Per-connection buffer / timeout knobs consulted on the uWS thread for the lifetime of the server. Every field is optional; unset fields fall back to the defaults resolved inside ws_server.cpp at the use site.	20
Identity	Identity of the connected client. Set once on connect by the dispatcher from the WebSocket upgrade authenticator. Read-only externally. Other components / admin tools can SELECT * FROM local.jsonrpc to enumerate currently-connected clients.	20
NamedSelection	Named arm of MemberSelector: matches only the listed members. Wire shape: `{type: "NamedSelection", value: {memberNames: [...]}}`.	20
ObjectInfo	One entry of listObjects' result.	20
ObjectState	Per-object snapshot returned by `getObjectsBatchState`. `properties` carries successful reads (one entry per property name resolved against the object's class hierarchy); `errors` is empty when nothing failed for this object.	21
PropertyError	Per-property read error. Used when the caller-supplied `propertyNames` includes a name that doesn't exist on a given object's class, or when the accessor itself throws. Carried in a separate list from successful reads so a property literally named `error` can never collide with the failure envelope.	21
PropertyValuePair	Per-property changed value. The value type is decided by the addressed kernel object, not by JSON-RPC, so STL can't statically describe it.	21
RegisteredBundleInfo	Summary record returned by `listRegisteredBundles`. Excludes the file bytes; just enough to answer "what is currently being served, and where".	21
SessionInfo	One discovered session and the buses currently detected inside it. Names are opaque strings from the kernel's session-discovery surface.	21
StaticBundle	One bundle of in-memory files served on a configurable URL prefix. The same bundle's files are served at `<urlPrefix>/<file.path>`. `indexFileName` is the SPA fallback served whenever a request under `urlPrefix` does not match any file (lets the client-side router handle paths the server does not know about).	22
StaticFile	One file inside a StaticBundle. Bytes are owned by the bundle; `path` is relative to the bundle's `urlPrefix` and uses forward slashes regardless of the producer's OS. `contentType` is the MIME type the server will return verbatim; the bundle producer decides it (no extension-based inference happens server-side).	22
SubscribeBlock	Pre-subscription block bundled into createInterest. Wires the listed property / event subscriptions on every matched object -- including ones that arrive later. Wire shape mirrors the typed STL representation exactly: each `MaybeMemberSelector` is either omitted (no auto-subscribe), `null`, or one of the typed variant arms (`{type: "WildcardSelection", value: {}}` or `{type: "NamedSelection", value: {memberNames: [...]}}`); `maxRateHz` is an optional positive number. Strict alignment with the typed surface lets the dispatcher's uniform meta-dispatch path consume the wire directly via `fromJson`, without a per-method translation step.	22
TlsConfig	TLS materials. The schema is reserved so configs stay forward-compatible, but the component fails to start if `tls` is set; full TLS support is not yet wired through the uWebSockets SSL backend.	22
TypeLookupResult	Bundled return of getType: always the CustomTypeSpec, plus the JSON-Schema fragment when the caller opted in. `schema` is empty when not requested.	22
WildcardSelection	Wildcard arm of MemberSelector: matches every member of the relevant kind. Wire shape: `{type: "WildcardSelection", value: {}}`.	23

`AddedObjectEntry` STRUCTURE

One object newly entering an interest's match set.

Name	Type	Description
`objectName`	string	name of the newly-matched object
`qualifiedClassName`	string	package-qualified class name of the object
`currentValues`	PropertyValueList	initial values of pre-subscribed properties (empty if none wired)

Named by AddedObjectList.

Configuration STRUCTURE

JSON-RPC over WebSocket component configuration. Only `address` and `port` are required; every other field falls back to a documented default.

Name	Type	Description
`address`	string	The IP address or hostname the server will bind to
`port`	u16	The port number the server will listen on
`updateFreqHz`	f64	Run-loop frequency in Hz; defaults to 60
`connectionLimits`	ConnectionLimits	Per-connection uWS buffer / timeout knobs
`tls`	TlsConfig	schema placeholder; setting this fails component start
`controlBusName`	string	Control-plane bus on `local`; defaults to "jsonrpc_control"

ConnectionLimits STRUCTURE

Per-connection buffer / timeout knobs consulted on the uWS thread for the lifetime of the server. Every field is optional; unset fields fall back to the defaults resolved inside ws_server.cpp at the use site.

Name	Type	Description
`idleTimeoutSeconds`	u16	uWS idle close timeout; defaults to 120
`maxPayloadBytes`	u32	Max inbound frame size; defaults to 16 KiB
`maxBackpressureBytes`	u32	uWS hard outbound buffer ceiling; defaults to 64 KiB
`highBackpressureBytes`	u32	Soft trigger for the BackpressureUpdate signal; defaults to 32 KiB

Named by Configuration.

Identity STRUCTURE

Identity of the connected client. Set once on connect by the dispatcher from the WebSocket upgrade authenticator. Read-only externally. Other components / admin tools can SELECT * FROM local.jsonrpc to enumerate currently-connected clients.

Name	Type	Description
`subject`	string	authenticated subject identifier (e.g. user / service principal)

Named by JsonRpcServer.

NamedSelection STRUCTURE

Named arm of MemberSelector: matches only the listed members. Wire shape: `{type: "NamedSelection", value: {memberNames: [...]}}`.

Name	Type	Description
`memberNames`	sen.kernel.StringList	member names to match

Named by MemberSelector.

ObjectInfo STRUCTURE

One entry of listObjects' result.

Name	Type	Description
`objectName`	string	object name within the interest's match set
`qualifiedClassName`	string	package-qualified class name of the object

Named by ObjectInfos.

`ObjectState` STRUCTURE

Per-object snapshot returned by `getObjectsBatchState`. `properties` carries successful reads (one entry per property name resolved against the object's class hierarchy); `errors` is empty when nothing failed for this object.

Name	Type	Description
`objectName`	string	object name within the interest's match set
`qualifiedClassName`	string	package-qualified class name at read time
`properties`	PropertyValueList	resolved property values; empty when caller's filter matched nothing
`errors`	PropertyErrorList	per-property failures; empty when none occurred

Named by ObjectStateList.

`PropertyError` STRUCTURE

Per-property read error. Used when the caller-supplied `propertyNames` includes a name that doesn't exist on a given object's class, or when the accessor itself throws. Carried in a separate list from successful reads so a property literally named `error` can never collide with the failure envelope.

Name	Type	Description
`propertyName`	string	name from the caller's `propertyNames` list that could not be read
`error`	string	human-readable reason (e.g. "unknown property")

Named by PropertyErrorList.

`PropertyValuePair` STRUCTURE

Per-property changed value. The value type is decided by the addressed kernel object, not by JSON-RPC, so STL can't statically describe it.

Name	Type	Description
`propertyName`	string	property member name
`value`	string	JSON-encoded SenValue

Named by PropertyValueList.

`RegisteredBundleInfo` STRUCTURE

Summary record returned by `listRegisteredBundles`. Excludes the file bytes; just enough to answer "what is currently being served, and where".

Name	Type	Description
`bundleId`	u64	opaque id assigned by the server, stable for the bundle's lifetime
`urlPrefix`	string	bundle's URL prefix
`fileCount`	u64	number of files in the bundle

Named by RegisteredBundleInfoList.

`SessionInfo` STRUCTURE

One discovered session and the buses currently detected inside it. Names are opaque strings from the kernel's session-discovery surface.

Name	Type	Description
`name`	string	session name
`buses`	sen.kernel.StringList	bus names currently detected in this session

Named by SessionInfoList.

StaticBundle STRUCTURE

One bundle of in-memory files served on a configurable URL prefix. The same bundle's files are served at `<urlPrefix>/<file.path>`. `indexFileName` is the SPA fallback served whenever a request under `urlPrefix` does not match any file (lets the client-side router handle paths the server does not know about).

Name	Type	Description
`urlPrefix`	string	URL prefix, e.g. "/explorer"; must start with "/"
`indexFileName`	string	SPA fallback, e.g. "index.html"; must match a `path` in `files`
`files`	StaticFileList	file set

Named by StaticFileServer.

StaticFile STRUCTURE

One file inside a StaticBundle. Bytes are owned by the bundle; `path` is relative to the bundle's `urlPrefix` and uses forward slashes regardless of the producer's OS. `contentType` is the MIME type the server will return verbatim; the bundle producer decides it (no extension-based inference happens server-side).

Name	Type	Description
`path`	string	path within the bundle, e.g. "assets/main.js"
`contentType`	string	MIME type, e.g. "text/html"
`contents`	sen.kernel.Buffer	raw file bytes

Named by StaticFileList.

SubscribeBlock STRUCTURE

Pre-subscription block bundled into createInterest. Wires the listed property / event subscriptions on every matched object -- including ones that arrive later. Wire shape mirrors the typed STL representation exactly: each `MaybeMemberSelector` is either omitted (no auto-subscribe), `null`, or one of the typed variant arms (`{type: "WildcardSelection", value: {}}` or `{type: "NamedSelection", value: {memberNames: [...]}}`); `maxRateHz` is an optional positive number. Strict alignment with the typed surface lets the dispatcher's uniform meta-dispatch path consume the wire directly via `fromJson`, without a per-method translation step.

Name	Type	Description
`properties`	MemberSelector	which properties to auto-subscribe to (or none)
`events`	MemberSelector	which events to auto-subscribe to (or none)
`maxRateHz`	f64	optional throttle cap shared across the wired subscriptions

Named by JsonRpcServer.

TlsConfig STRUCTURE

TLS materials. The schema is reserved so configs stay forward-compatible, but the component fails to start if `tls` is set; full TLS support is not yet wired through the uWebSockets SSL backend.

Name	Type	Description
`certPath`	string	PEM-encoded server certificate
`keyPath`	string	PEM-encoded server private key

Named by Configuration.

TypeLookupResult STRUCTURE

Bundled return of getType: always the CustomTypeSpec, plus the JSON-Schema fragment when the caller opted in. `schema` is empty when not requested.

Name	Type	Description
`spec`	sen.kernel.CustomTypeSpec	

Name	Type	Description
`schema`	string	JSON-encoded schema fragment; "" when withSchema=false

Named by JsonRpcServer.

`WildcardSelection` STRUCTURE

Wildcard arm of MemberSelector: matches every member of the relevant kind. Wire shape: `{type: "WildcardSelection", value: {}}`.

Named by MemberSelector.

Variant records

MemberSelector	Selector for properties or events in a SubscribeBlock. Absent = no auto-subscribe; WildcardSelection = "*"; NamedSelection = explicit list.	23

`MemberSelector` VARIANT

Selector for properties or events in a SubscribeBlock. Absent = no auto-subscribe; WildcardSelection = "*"; NamedSelection = explicit list.

Type	Description
WildcardSelection	
NamedSelection	

Named by SubscribeBlock.

Arrays

AddedObjectList	23
ObjectInfos	23
ObjectStateList	23
PropertyErrorList	23
PropertyValueList	24
RegisteredBundleInfoList	24
SessionInfoList	24
StaticFileList	24

`AddedObjectList` SEQUENCE

```
sequence<AddedObjectEntry> AddedObjectList;
```

Named by JsonRpcServer.

`ObjectInfos` SEQUENCE

```
sequence<ObjectInfo> ObjectInfos;
```

Named by JsonRpcServer.

`ObjectStateList` SEQUENCE

```
sequence<ObjectState> ObjectStateList;
```

Named by JsonRpcServer.

`PropertyErrorList` SEQUENCE

```
sequence<PropertyError> PropertyErrorList;
```

Named by ObjectState.

PropertyValueList SEQUENCE

```
sequence<PropertyValuePair> PropertyValueList;
```

Named by AddedObjectEntry, JsonRpcServer, ObjectState.

RegisteredBundleInfoList SEQUENCE

```
sequence<RegisteredBundleInfo> RegisteredBundleInfoList;
```

Named by StaticFileServer.

SessionInfoList SEQUENCE

```
sequence<SessionInfo> SessionInfoList;
```

Named by JsonRpcServer.

StaticFileList SEQUENCE

```
sequence<StaticFile> StaticFileList;
```

Named by StaticBundle.

Optionals

MaybeConnectionLimits		24
MaybeControlBusName		24
MaybeFlag		24
MaybeHighBackpressureBytes		24
MaybeIdleTimeoutSeconds		25
MaybeMaxBackpressureBytes		25
MaybeMaxPayloadBytes		25
MaybeMemberSelector		25
MaybeRateHz		25
MaybeStringList	Optional list of object names. Absent on the wire (or `null`) selects every object currently in the addressed interest's match set; present limits the read to the listed names.	25
MaybeSubscribeBlock		25
MaybeTlsConfig		25
MaybeUpdateFreqHz		25

MaybeConnectionLimits OPTIONAL

```
optional<ConnectionLimits> MaybeConnectionLimits;
```

MaybeControlBusName OPTIONAL

```
optional<string> MaybeControlBusName;
```

MaybeFlag OPTIONAL

```
optional<bool> MaybeFlag;
```

MaybeHighBackpressureBytes OPTIONAL

```
optional<u32> MaybeHighBackpressureBytes;
```

MaybeIdleTimeoutSeconds OPTIONAL

```
optional<u16> MaybeIdleTimeoutSeconds;
```

MaybeMaxBackpressureBytes OPTIONAL

```
optional<u32> MaybeMaxBackpressureBytes;
```

MaybeMaxPayloadBytes OPTIONAL

```
optional<u32> MaybeMaxPayloadBytes;
```

MaybeMemberSelector OPTIONAL

```
optional<MemberSelector> MaybeMemberSelector;
```

MaybeRateHz OPTIONAL

```
optional<f64> MaybeRateHz;
```

MaybeStringList OPTIONAL

Optional list of object names. Absent on the wire (or `null`) selects every object currently in the addressed interest's match set; present limits the read to the listed names.

```
optional<sen.kernel.StringList> MaybeStringList;
```

MaybeSubscribeBlock OPTIONAL

```
optional<SubscribeBlock> MaybeSubscribeBlock;
```

MaybeTlsConfig OPTIONAL

```
optional<TlsConfig> MaybeTlsConfig;
```

MaybeUpdateFreqHz OPTIONAL

```
optional<f64> MaybeUpdateFreqHz;
```

logmaster

3 types.

Object classes

LogMaster	allows controlling the loggers	26
Logger	represents a logger	26

`LogMaster` CLASS

allows controlling the loggers

Name	Type	Flags	Description
`targetBus`	string	RO S BE	

Method	Signature	Description
`muteAll`	`()`	mute all the loggers
`unmuteAll`	`()`	un-mutes all the muted loggers
`toggleMuteAll`	`()`	toggles the muting of all loggers
`setLevel`	`(level: sen.kernel.log.LogLevel)`	sets all the loggers to a given level

`Logger` CLASS

represents a logger

Name	Type	Flags	Description
`level`	sen.kernel.log.LogLevel	RO D BE	the current log level of this logger
`pattern`	string	RO D C	the pattern used by this logger
`muted`	bool	RO D BE	true if muted

Method	Signature	Description
`mute`	`()`	mute this logger
`unmute`	`()`	unmute this logger
`setLevel`	`(level: sen.kernel.log.LogLevel)`	unmute this logger
`setPattern`	`(pattern: string)`	sets the pattern for this logger

Fixed records

Config	component configuration	26

`Config` STRUCTURE

component configuration

Name	Type	Description
`targetBus`	string	defaults to "local.log"
`period`	sen.Duration	defaults to 500 milliseconds

py

3 types.

Object classes

PythonInterpreter	represents the embedded Python interpreter	27

PythonInterpreter CLASS

represents the embedded Python interpreter

Method	Signature	Description
`eval`	`(expr: string) → string`	evaluates a Python expression
`exec`	`(code: string)`	executes a Python code block
`execFile`	`(file: string)`	executes a Python file

Fixed records

Configuration	Overall component configuration	27

Configuration STRUCTURE

Overall component configuration

Name	Type	Description
`freqHz`	f32	update frequency in Hertz
`module`	string	python module to use (name of a python script, without the .py)
`imports`	StringList	sen packages to import
`bus`	string	bus where to publish the interpreter object (none by default)

Arrays

StringList	27

StringList SEQUENCE

```
sequence<string> StringList;
```

Named by Configuration.

recorder

6 types.

Object classes

Recorder	represents an recording session	28

`Recorder` CLASS

represents an recording session

Name	Type	Flags	Description
`settings`	RecordingSettings	RO S BE	
`state`	RecorderState	RO D C	

Method	Signature	Description
`start`	`()`	starts (or resumes) recording
`stop`	`()`	stops recording
`fetchStats`	`() → sen.db.OutStats`	sample the output statistics

Fixed records

Configuration	Overall component configuration	28
RecordingSettings	Configuration of a predefined recording	28

`Configuration` STRUCTURE

Overall component configuration

Name	Type	Description
`samplingPeriod`	sen.Duration	sampling period for the recordings
`recordings`	RecordingSettingsList	predefined recordings
`bus`	string	bus to publish the object (local.rec by default)

`RecordingSettings` STRUCTURE

Configuration of a predefined recording

Name	Type	Description
`name`	string	name of the recording (cannot be empty)
`folder`	string	storage folder
`selections`	SelectionList	which objects shall be recorded
`keyframePeriod`	sen.Duration	period between keyframes (0 means no keyframes)
`indexKeyframes`	bool	should we create indexes for keyframes
`indexObjects`	bool	should we create indexes for objects
`autoStart`	bool	should we start recording automatically

Named by Recorder, RecordingSettingsList.

Enumerations

RecorderState	the state of a recording session	28

`RecorderState` ENUMERATION

the state of a recording session

Held as u8.

stopped	0	error	2
recording	1		

Named by Recorder.

Arrays

RecordingSettingsList	List of predefined recordings	29
SelectionList	List of selection queries	29

`RecordingSettingsList` **SEQUENCE**

List of predefined recordings

```
sequence<RecordingSettings> RecordingSettingsList;
```

Named by Configuration.

`SelectionList` **SEQUENCE**

List of selection queries

```
sequence<string> SelectionList;
```

Named by RecordingSettings.

replayer

4 types.

Object classes

Replay	controls the replay of a given archive	30
Replayer	represents the replayer component and manages Replay objects.	30

`Replay` CLASS

controls the replay of a given archive

Name	Type	Flags	Description
`archiveInfo`	sen.db.Summary	RO S BE	information about the archive being replayed
`archivePath`	string	RO S BE	file path to the archive
`status`	ReplayStatus	RO D C	playback status
`playbackTime`	sen.TimeStamp	RO D BE	playback time

Method	Signature	Description
`play`	`()`	starts the playback from the current position
`pause`	`()`	hold the playback position, if playing
`stop`	`()`	goes to the start of the playback, erasing all objects
`seek`	`(time: sen.TimeStamp)`	restores the playback to the keyframe that's closest to a given time
`advance`	`(time: sen.Duration)`	if paused or stopped, advances the playback a given time delta

`Replayer` CLASS

represents the replayer component and manages Replay objects.

Name	Type	Flags	Description
`autoPlay`	bool	RO S BE	if true, automatically calls play when opening a replay

Method	Signature	Description
`open`	`(name: string, path: string)`	opens an archive in a given path, and creates a named Replay object.
`close`	`(name: string)`	closes the named Replay object, if any.
`closeAll`	`()`	closes all previously opened replays

Fixed records

Configuration	Overall component configuration	30

`Configuration` STRUCTURE

Overall component configuration

Name	Type	Description
`bus`	string	bus to publish the object (local.replay by default)
`samplingPeriod`	sen.Duration	max sampling period for flushing the replayed data
`autoOpen`	string	optional archive path to auto open on start-up (empty by default)
`autoPlay`	bool	if autoOpen is not empty, try to play it on start-up (false by default)

Enumerations

ReplayStatus the state in which a Replay can be 31

`ReplayStatus` ENUMERATION

the state in which a Replay can be

Held as `u8`.

stopped	0
paused	1
playing	2

Named by Replay.

rest

37 types.

Fixed records

Argument 32
Bus 32
BusSummary 32
ClientAuth 32
Configuration 33
Error 33
Interest 33
InterestSummary 33
Link 33
Notification 33
Object 33
ObjectEvent 34
ObjectMethod 34
ObjectSummary 34
Session 34
SessionSummary 34
SubscriptionOptions 34
Subscriptions 34
Success 35
Version 35

`Argument` STRUCTURE

Name	Type	Description
`name`	string	
`description`	string	
`type`	string	

Named by Arguments.

`Bus` STRUCTURE

Name	Type	Description
`id`	string	
`status`	Status	
`objects`	ObjectsSummary	

`BusSummary` STRUCTURE

Name	Type	Description
`id`	string	
`status`	Status	

Named by Buses.

`ClientAuth` STRUCTURE

Name	Type	Description
`id`	string	

`Configuration` STRUCTURE

Name	Type	Description
`address`	string	The IP address or hostname the server will bind to
`port`	u16	The port number the server will listen on
`freqHz`	f32	Update frequency in Hertz

`Error` STRUCTURE

Name	Type	Description
`error`	string	

`Interest` STRUCTURE

Name	Type	Description
`query`	string	
`name`	string	

`InterestSummary` STRUCTURE

Name	Type	Description
`name`	string	
`session`	string	
`bus`	string	

Named by InterestsSummary.

`Link` STRUCTURE

Name	Type	Description
`rel`	RelType	
`href`	string	
`method`	HttpMethod	

Named by Links, ObjectSummary.

`Notification` STRUCTURE

Name	Type	Description
`type`	NotificationType	
`interest`	string	
`time`	sen.TimeStamp	
`data`	string	

`Object` STRUCTURE

Name	Type	Description
`objectId`	u32	
`name`	string	
`className`	string	
`localName`	string	

Name	Type	Description
`description`	string	
`links`	Links	

`ObjectEvent` STRUCTURE

Name	Type	Description
`name`	string	
`description`	string	

`ObjectMethod` STRUCTURE

Name	Type	Description
`name`	string	
`description`	string	
`retType`	string	
`args`	Arguments	

`ObjectSummary` STRUCTURE

Name	Type	Description
`objectId`	u32	
`name`	string	
`className`	string	
`localName`	string	
`link`	Link	

Named by ObjectsSummary.

`Session` STRUCTURE

Name	Type	Description
`name`	string	

`SessionSummary` STRUCTURE

Name	Type	Description
`name`	string	
`buses`	BusNames	

`SubscriptionOptions` STRUCTURE

Name	Type	Description
`maxUpdateTime`	sen.Duration	Max. update time

`Subscriptions` STRUCTURE

Name	Type	Description
`properties`	Properties	
`events`	Events	

`Success` STRUCTURE

Name	Type	Description
`msg`	string	

`Version` STRUCTURE

Name	Type	Description
`version`	string	

Enumerations

HttpMethod 35
InvokeStatus 35
NotificationType 35
RelType 35
Status 36

`HttpMethod` ENUMERATION

Held as u8.

httpGet	0
httpPost	1
httpPut	2
httpDelete	3
httpOption	4

Named by Link.

`InvokeStatus` ENUMERATION

Held as u8.

pending	0
finished	1
failed	2

`NotificationType` ENUMERATION

Held as u8.

evt	0
invoke	1
property	2
objectAdded	3
objectRemoved	4
count	5

Named by Notification.

`RelType` ENUMERATION

Held as u8.

method	0
setter	1
getter	2
evt	3
property	4
propertySubscribe	5
propertyUnsubscribe	6
def	7

Named by Link.

Status ENUMERATION

Held as u8.

open	0
close	1

Named by Bus, BusSummary.

Arrays

Arguments 36
BusNames 36
Buses 36
Events 36
InterestsSummary 36
Links 36
ObjectsSummary 36
Properties 37
Sessions 37

Arguments SEQUENCE

```
sequence<Argument> Arguments;
```

Named by ObjectMethod.

BusNames SEQUENCE

```
sequence<string> BusNames;
```

Named by SessionSummary.

Buses SEQUENCE

```
sequence<BusSummary> Buses;
```

Events SEQUENCE

```
sequence<string> Events;
```

Named by Subscriptions.

InterestsSummary SEQUENCE

```
sequence<InterestSummary> InterestsSummary;
```

Links SEQUENCE

```
sequence<Link> Links;
```

Named by Object.

ObjectsSummary SEQUENCE

```
sequence<ObjectSummary> ObjectsSummary;
```

Named by Bus.

Properties SEQUENCE

```
sequence<string> Properties;
```

Named by Subscriptions.

Sessions SEQUENCE

```
sequence<string> Sessions;
```

Optionals

ThreadPoolSize 37
UpdateFrequency 37
UpdateTime 37

ThreadPoolSize OPTIONAL

```
optional<u16> ThreadPoolSize;
```

UpdateFrequency OPTIONAL

```
optional<f32> UpdateFrequency;
```

UpdateTime OPTIONAL

```
optional<sen.Duration> UpdateTime;
```

shell

27 types.

Object classes

Shell	Provides a text-based interface to interact with the a sen kernel	38

Shell CLASS

Provides a text-based interface to interact with the a sen kernel

Name	Type	Flags	Description
`config`	Configuration	RO S BE	service configuration

Method	Signature	Description
`clear`	`()`	clears the screen
`help`	`()`	basic usage help
`history`	`()`	lists previous commands
`ls`	`()`	lists all the existing objects and sources.
`open`	`(source: string)`	opens a source (bus or session) without any condition
`query`	`(name: string, condition: string)`	opens a named source using a condition expression
`close`	`(source: string)`	closes a previously-opened source (session, bus or query)
`info`	`(term: string)`	prints information about a type or instance
`shutdown`	`()`	requests the kernel to be shut-down
`src`	`()`	lists the current sources

Fixed records

CPrint		38
ClearCurrentLine		39
ClearRemainingCurrentLine		39
ClearScreen		39
Configuration	Configuration params for the shell	39
HideCursor		39
MoveCursorAllLeft		39
MoveCursorDown		39
MoveCursorLeft		39
MoveCursorRight		39
MoveCursorUp		40
NewLine		40
Print		40
Query	Holds information about a named query	40
RestoreCursorPosition		40
SaveCursorPosition		40
SetBgColor		40
SetFgColor		40
SetWindowTitle		40
ShowCursor		41

CPrint STRUCTURE

Name	Type	Description
`flags`	u32	
`color`	u32	
`text`	string	

Named by TerminalCmd.

`ClearCurrentLine` STRUCTURE

Named by TerminalCmd.

`ClearRemainingCurrentLine` STRUCTURE

Named by TerminalCmd.

`ClearScreen` STRUCTURE

Named by TerminalCmd.

`Configuration` STRUCTURE

Configuration params for the shell

Name	Type	Description
`prompt`	string	defaults to 'sen> '
`timeStyle`	TimeStyle	how to display time
`bufferStyle`	BufferStyle	how to display buffers
`serverEnabled`	bool	enables remote shell access (false by default)
`serverPort`	u16	0 means default
`noLogo`	bool	true if you don't want a welcoming logo
`open`	SourcesList	list of sources to open automatically
`query`	QueryList	list of queries to create automatically
`logBus`	string	where to find the logmaster, if any (default: local.log)

Named by Shell.

`HideCursor` STRUCTURE

Named by TerminalCmd.

`MoveCursorAllLeft` STRUCTURE

Named by TerminalCmd.

`MoveCursorDown` STRUCTURE

Name	Type	Description
`cells`	u32	

Named by TerminalCmd.

`MoveCursorLeft` STRUCTURE

Name	Type	Description
`cells`	u32	

Named by TerminalCmd.

`MoveCursorRight` STRUCTURE

Name	Type	Description
`cells`	u32	

Named by TerminalCmd.

`MoveCursorUp` STRUCTURE

Name	Type	Description
`cells`	u32	

Named by TerminalCmd.

`NewLine` STRUCTURE

Named by TerminalCmd.

`Print` STRUCTURE

Name	Type	Description
`text`	string	

Named by TerminalCmd.

`Query` STRUCTURE

Holds information about a named query

Name	Type	Description
`name`	string	the name of the query (cannot contain spaces or dots)
`selection`	string	the query expression

Named by QueryList.

`RestoreCursorPosition` STRUCTURE

Named by TerminalCmd.

`SaveCursorPosition` STRUCTURE

Named by TerminalCmd.

`SetBgColor` STRUCTURE

Name	Type	Description
`color`	u32	

Named by TerminalCmd.

`SetFgColor` STRUCTURE

Name	Type	Description
`color`	u32	

Named by TerminalCmd.

`SetWindowTitle` STRUCTURE

Name	Type	Description
`text`	string	

Named by TerminalCmd.

ShowCursor STRUCTURE

Named by TerminalCmd.

Enumerations

BufferStyle	How buffers should be printed	41
ListStyle	How lists should be printed	41
TimeStyle	How timestamps should be printed	41

BufferStyle ENUMERATION

How buffers should be printed

Held as u8.

size	0
hexdump	1

Named by Configuration.

ListStyle ENUMERATION

How lists should be printed

Held as u8.

treeStyle	0
plainStyle	1

TimeStyle ENUMERATION

How timestamps should be printed

Held as u8.

timestampUTC	0
timestampLocal	1

Named by Configuration.

Variant records

TerminalCmd	41

TerminalCmd VARIANT

Type	Description
Print	
CPrint	
NewLine	
MoveCursorLeft	
MoveCursorRight	
MoveCursorUp	
MoveCursorDown	
MoveCursorAllLeft	
SetFgColor	
SetBgColor	
ClearCurrentLine	
ClearRemainingCurrentLine	
HideCursor	
ShowCursor	
SaveCursorPosition	

Type	Description
RestoreCursorPosition	
ClearScreen	
SetWindowTitle	

Arrays

QueryList 42
SourcesList 42

`QueryList` SEQUENCE

```
sequence<Query> QueryList;
```

Named by Configuration.

`SourcesList` SEQUENCE

```
sequence<string> SourcesList;
```

Named by Configuration.

term

8 types.

Object classes

Term Modern TUI-based interactive terminal for the sen kernel 43

`Term` CLASS

Modern TUI-based interactive terminal for the sen kernel

Name	Type	Flags	Description
`config`	Configuration	RO S BE	service configuration

Fixed records

Configuration Configuration params for the term component 43
QueryDef Holds information about a named query 43

`Configuration` STRUCTURE

Configuration params for the term component

Name	Type	Description
`initialScope`	string	initial scope, e.g. "local.demo"; its bus must be in open too. Empty means root.
`timeStyle`	TimeStyle	how to display time (default: utc)
`theme`	ThemeStyle	color theme (default: catppuccinMocha)
`noLogo`	bool	true to suppress the welcome banner
`open`	SourcesList	sources to open automatically at startup
`query`	QueryList	named queries to create automatically at startup
`listen`	NameList	event listeners to register at startup (object or object.event)
`callTimeout`	sen.Duration	async call timeout (default: 30s)

Named by Term.

`QueryDef` STRUCTURE

Holds information about a named query

Name	Type	Description
`name`	string	the name of the query (cannot contain spaces or dots)
`selection`	string	the query expression

Named by QueryList.

Enumerations

ThemeStyle Color theme for the terminal UI 43
TimeStyle How timestamps should be displayed 44

`ThemeStyle` ENUMERATION

Color theme for the terminal UI

Held as u8.

oneDark	0	gruvboxDark	6
oneLight	1	gruvboxLight	7
catppuccinMocha	2	tokyoNight	8
catppuccinLatte	3	solarizedLight	9
dracula	4		
nord	5		

Named by Configuration.

TimeStyle ENUMERATION

How timestamps should be displayed

Held as u8.

utc	0
local	1

Named by Configuration.

Arrays

NameList 44
QueryList 44
SourcesList 44

NameList SEQUENCE

```
sequence<string> NameList;
```

Named by Configuration.

QueryList SEQUENCE

```
sequence<QueryDef> QueryList;
```

Named by Configuration.

SourcesList SEQUENCE

```
sequence<string> SourcesList;
```

Named by Configuration.

db

3 types.

Fixed records

OutSettings	Settings for outputs	45
OutStats	Statistics of writing	45
Summary	Summary of an archive	45

`OutSettings` STRUCTURE

Settings for outputs

Name	Type	Description
`name`	string	name of the archive (cannot be empty)
`folder`	string	storage folder
`indexKeyframes`	bool	should we create indexes for keyframes

`OutStats` STRUCTURE

Statistics of writing

Name	Type	Description
`maxQueueSize`	u64	in bytes

Named by sen.components.recorder.Recorder.

`Summary` STRUCTURE

Summary of an archive

Name	Type	Description
`firstTime`	sen.TimeStamp	the time of the first event
`lastTime`	sen.TimeStamp	the time of the last event
`keyframeCount`	u32	number of written keyframes
`objectCount`	u32	number of recorded objects
`typeCount`	u32	number of types written
`annotationCount`	u32	number of annotations
`indexedObjectCount`	u32	number of indexed objects

Named by sen.components.replayer.Replay.

kernel

160 types.

Object classes

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VirtualMasterClock	Controls all the VirtualKernelClock that it discovers in the bus where it's published. One step reaches the next component due to run: each clock reports how long until its own next one, and the soonest wins	46

`KernelApi` CLASS

Represents the kernel

Name	Type	Flags	Description
`buildInfo`	BuildInfo	RO S BE	kernel build information

Method	Signature	Description
`shutdown`	`()`	requests the kernel to shut down
`getUnits`	`() → UnitList`	registered units
`getTypes`	`() → StringList`	registered types, by name
`getConfig`	`() → KernelParams`	configuration of the kernel execution
`getRuntimeNetworkFootprint`	`() → NetworkFootprint`	actual network addresses and ports currently used by this process

---

`VirtualClock` CLASS

Allows inspecting and controlling the advance of virtual time and associated component iteration

Name	Type	Flags	Description
`time`	sen.TimeStamp	RO D BE	the current virtual time

`VirtualKernelClock` CLASS

A clock for all the components loaded by the kernel

Method	Signature	Description
`processNoFlush`	`(delta: sen.Duration) → sen.Duration`	updates the time, drains the inputs, and cycles components - doesn't flush outputs
`flushOutputs`	`() → sen.Duration`	makes component outputs visible to other components and returns the next step duration

---

`VirtualMasterClock` CLASS

Controls all the VirtualKernelClock that it discovers in the bus where it's published. One step reaches the next component due to run: each clock reports how long until its own next one, and the soonest wins

Method	Signature	Description
`advanceTime`	`(duration: sen.Duration) → sen.Duration`	steps until the virtual time advanced reaches duration, rounding up to a whole step. Returns the wall time the call took
`step`	`() → sen.Duration`	single step forward
`steps`	`(count: u64) → sen.Duration`	perform 'count' steps forward

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AliasTypeSpec STRUCTURE

data of an aliased type

Name	Type	Description
`aliasedType`	string	qualified type name being aliased

Named by CustomTypeData.

AliasTypeSpecV4 STRUCTURE

Name	Type	Description
`aliasedType`	string	qualified type name being aliased

Named by CustomTypeDataV4.

AliasTypeSpecV5 STRUCTURE

Name	Type	Description
`aliasedType`	string	qualified type name being aliased
`hash`	u32	unique hash

Named by CustomTypeDataV5.

ArgSpec STRUCTURE

an argument of a callable (methods and events)

Name	Type	Description
`name`	string	argument name
`description`	string	documentation
`type`	string	qualified type name of the argument

Named by ArgSpecList.

ArgSpecV4 STRUCTURE

Name	Type	Description
`name`	string	argument name
`description`	string	documentation
`type`	string	qualified type name of the argument

Named by ArgSpecListV4.

`ArgSpecV5` STRUCTURE

Name	Type	Description
`name`	string	argument name
`nameHash`	u32	hash computed with the argument name
`description`	string	documentation
`type`	string	qualified type name of the argument
`hash`	u32	unique hash

Named by ArgSpecListV5.

`BuildInfo` STRUCTURE

Build-related information

Name	Type	Description
`maintainer`	string	principal maintainer of the software
`version`	string	version string (format-agnostic)
`compiler`	string	vendor-specific compiler string
`debugMode`	bool	compiled in debug mode or not
`buildTime`	string	the commit's timestamp, ISO-8601; not the compiler's clock
`wordSize`	WordSize	architecture
`gitRef`	string	git ref spec
`gitHash`	string	git hash
`gitStatus`	GitStatus	git status

Named by ComponentInfo, KernelApi.

`BuiltComponentParams` STRUCTURE

Parameters related to components build by the kernel

Name	Type	Description
`name`	string	the name of the component
`period`	sen.Duration	the execution period
`config`	ComponentConfig	the configuration of the component
`imports`	StringList	the imported packages

Named by BuiltComponentList.

`BusAddress` STRUCTURE

Address to connect to a bus

Name	Type	Description
`sessionName`	string	
`busName`	string	

`ClassTypeSpec` STRUCTURE

spec of a class

Name	Type	Description
`properties`	PropertySpecList	
`methods`	MethodSpecList	
`events`	EventSpecList	
`constructor`	MethodSpec	

Name	Type	Description
`parents`	StringList	
`isInterface`	bool	

Named by CustomTypeData.

`ClassTypeSpecV4` STRUCTURE

Name	Type	Description
`properties`	PropertySpecListV4	
`methods`	MethodSpecListV4	
`events`	EventSpecListV4	
`isInterface`	bool	
`id`	u32	
`constructor`	MethodSpecV4	
`parents`	StringList	

Named by CustomTypeDataV4.

`ClassTypeSpecV5` STRUCTURE

Name	Type	Description
`properties`	PropertySpecListV5	
`methods`	MethodSpecListV5	
`events`	EventSpecListV5	
`isInterface`	bool	
`id`	u32	
`constructor`	MethodSpecV5	
`parents`	StringList	
`hash`	u32	

Named by CustomTypeDataV5.

`ComponentConfig` STRUCTURE

Basic runtime configuration for a component

Name	Type	Description
`priority`	Priority	thread priority
`stackSize`	u32	thread stack size in bytes, 0 means default
`group`	u32	the group where to run the component
`cpuAffinity`	u64	CPU affinity mask
`inQueue`	QueueConfig	queuing of inbound information
`outQueue`	QueueConfig	queuing of outbound information
`sleepPolicy`	SleepPolicy	configurable component sleep policy

Named by BuiltComponentParams, LoadedComponentParams.

`ComponentInfo` STRUCTURE

Basic information of a kernel component

Name	Type	Description
`name`	string	
`description`	string	

Name	Type	Description
`buildInfo`	BuildInfo	

`CustomTypeSpec` STRUCTURE

spec of a custom type

Name	Type	Description
`name`	string	name of the type
`qualifiedName`	string	package-qualified name
`description`	string	documentation
`data`	CustomTypeData	spec data

Named by sen.components.jsonrpc.TypeLookupResult, CustomTypeSpecList.

`CustomTypeSpecV4` STRUCTURE

Name	Type	Description
`name`	string	name of the type
`qualifiedName`	string	package-qualified name
`description`	string	documentation
`data`	CustomTypeDataV4	spec data

Named by CustomTypeSpecListV4.

`CustomTypeSpecV5` STRUCTURE

Name	Type	Description
`name`	string	name of the type
`qualifiedName`	string	package-qualified name
`description`	string	documentation
`data`	CustomTypeDataV5	spec data

Named by CustomTypeSpecListV5.

`EnumTypeSpec` STRUCTURE

spec of an enum type

Name	Type	Description
`enums`	EnumeratorSpecList	enumerators
`storageType`	IntegralType	type used to store the value

Named by CustomTypeData.

`EnumTypeSpecV4` STRUCTURE

Name	Type	Description
`enums`	EnumeratorSpecListV4	enumerators
`storageType`	IntegralType	type used to store the value

Named by CustomTypeDataV4.

`EnumTypeSpecV5` STRUCTURE

Name	Type	Description
`enums`	EnumeratorSpecListV5	enumerators
`storageType`	IntegralType	type used to store the value
`hash`	u32	unique hash

Named by CustomTypeDataV5.

`EnumeratorSpec` STRUCTURE

an enumerator

Name	Type	Description
`name`	string	enumerator name
`key`	u32	enumerator value
`description`	string	enumerator description

Named by EnumeratorSpecList.

`EnumeratorSpecV4` STRUCTURE

Name	Type	Description
`name`	string	enumerator name
`key`	u64	enumerator value

Named by EnumeratorSpecListV4.

`EnumeratorSpecV5` STRUCTURE

Name	Type	Description
`name`	string	enumerator name
`key`	u64	enumerator value
`hash`	u32	unique hash

Named by EnumeratorSpecListV5.

`EnvVar` STRUCTURE

Information about an environment variable

Name	Type	Description
`name`	string	name of the variable
`value`	string	value of the variable

Named by EnvironmentVarList.

`ErrorData` STRUCTURE

Information related to the error

Name	Type	Description
`time`	sen.TimeStamp	time of the error
`errorMessage`	StringList	message presented to the user
`exceptionData`	UncaughtException	information about the unhandled exception (if any)
`signalData`	SignalData	information about the received signal (if any)

Named by ErrorReport.

`ErrorReport` STRUCTURE

Information about a detected error

Name	Type	Description
`senData`	SenData	information related to Sen
`processData`	ProcessData	information related to the process
`errorData`	ErrorData	information related to the error

`EventSpec` STRUCTURE

spec for an event

Name	Type	Description
`name`	string	name of the type
`description`	string	documentation
`args`	ArgSpecList	arguments
`transportMode`	TransportModeSpec	transport mode

Named by EventSpecList.

`EventSpecV4` STRUCTURE

Name	Type	Description
`memberHash`	u32	hash of the class member
`name`	string	name of the type
`description`	string	documentation
`args`	ArgSpecListV4	arguments
`transportMode`	TransportModeSpec	transport mode

Named by EventSpecListV4.

`EventSpecV5` STRUCTURE

Name	Type	Description
`memberHash`	u32	hash of the class member
`name`	string	name of the type
`description`	string	documentation
`args`	ArgSpecListV5	arguments
`transportMode`	TransportModeSpec	transport mode
`hash`	u32	unique hash

Named by EventSpecListV5.

`ExecError` STRUCTURE

Holds details of an execution error

Name	Type	Description
`category`	ErrorCategory	general problem
`explanation`	string	human-readable explanation

`KernelParams` STRUCTURE

This struct is reachable over the wire through Kernel.getConfig() and is embedded in SenData, and the encoding is positional, so adding a field here shifts what follows it either way.

Name	Type	Description
`runMode`	RunMode	how to run the kernel
`appName`	string	optional name of the application
`bus`	string	where to publish the kernel objects (defaults to local.kernel)
`clockBus`	string	where to publish the virtual clock (if any), defaults to bus
`clockName`	string	the name of the virtual clock (if any), defaults to 'clock'
`clockMaster`	bool	if time is virtualized, publish a master clock to clockBus
`logConfig`	log.Config	logging configuration
`crashReportDir`	string	where to store crash reports (defaults to the temp dir)
`crashReportDisabled`	bool	if true, no reports are generated
`lockMemoryPages`	bool	keep process pages memory-resident
`sleepPolicy`	SleepPolicy	configurable sleep policy of the kernel component
`compatibilityMode`	CompatibilityMode	what to do with a remote type that does not match
`limitCpuIdleLatency`	bool	keep the CPUs out of the deep idle states they are slow to leave

Named by KernelApi, SenData.

`LoadedComponentParams` STRUCTURE

Parameters related to a component that will be loaded by the kernel

Name	Type	Description
`name`	string	the name of the component
`config`	ComponentConfig	the configuration of the component

Named by LoadedComponentList.

`MethodSpec` STRUCTURE

spec for methods

Name	Type	Description
`name`	string	name of the type
`description`	string	documentation
`args`	ArgSpecList	arguments
`transportMode`	TransportModeSpec	transport mode
`constness`	MethodConstnessSpec	if it changes the state of the object
`deferred`	bool	if the execution of the method is deferred
`returnType`	string	qualified return type name
`propertyRelation`	PropertyRelationSpec	property relation of the method
`localOnly`	bool	true if the method can only be called locally

Named by ClassTypeSpec, MethodSpecList.

`MethodSpecV4` STRUCTURE

Name	Type	Description
`memberHash`	u32	hash of the class member
`name`	string	name of the type
`description`	string	documentation
`args`	ArgSpecListV4	arguments
`transportMode`	TransportModeSpec	transport mode

Name	Type	Description
`constness`	MethodConstness Spec	if it changes the state of the object
`deferred`	bool	true if the method is marked as deferred
`returnTypeId`	string	qualified return type name (empty for none)

Named by ClassTypeSpecV4, MethodSpecListV4.

MethodSpecV5 STRUCTURE

Name	Type	Description
`memberHash`	u32	hash of the class member
`name`	string	name of the type
`description`	string	documentation
`args`	ArgSpecListV5	arguments
`transportMode`	TransportModeSpec	transport mode
`constness`	MethodConstness Spec	if it changes the state of the object
`localOnly`	bool	true if the method can only be called locally
`returnTypeId`	string	qualified return type name (empty for none)
`hash`	u32	unique hash

Named by ClassTypeSpecV5, MethodSpecListV5.

NetworkFootprint STRUCTURE

Network endpoints predicted offline or observed at runtime for one process

Name	Type	Description
`discoveryPort`	u16	Discovery port used to calculate multicast addresses
`multicast`	NetworkFootprint Multicast	Multicast allocation details
`ports`	NetworkFootprintPort List	Port settings
`portExclusions`	NetworkFootprintPort Exclusions	Port ranges that cannot be used

Named by KernelApi.

NetworkFootprintAddressRange STRUCTURE

Inclusive IPv4 address range excluded from multicast allocation

Name	Type	Description
`min`	string	
`max`	string	

Named by NetworkFootprintAddressRangeList.

NetworkFootprintBus STRUCTURE

Multicast endpoint allocated to a bus

Name	Type	Description
`sessionName`	string	
`busName`	string	
`sessionId`	u32	

Name	Type	Description
`busId`	u32	
`groupAddress`	string	
`source`	NetworkFootprintBusSource	

Named by NetworkFootprintBusList.

`NetworkFootprintBusIdentity` STRUCTURE

Names and IDs that identify a bus

Name	Type	Description
`sessionName`	string	
`busName`	string	
`sessionId`	u32	
`busId`	u32	

Named by NetworkFootprintSelfCollision.

`NetworkFootprintCollision` STRUCTURE

Multicast allocation capacity and collision information

Name	Type	Description
`usableSpaceSize`	u64	Number of addresses available
`busCount`	u64	Number of reported buses
`probability`	f64	Estimated conflict from 0 to 1
`selfCollisions`	NetworkFootprintSelfCollisionList	Collisions in the reported buses

Named by NetworkFootprintMulticast.

`NetworkFootprintMulticast` STRUCTURE

Multicast information in the network footprint

Name	Type	Description
`port`	u16	Shared multicast port used by every bus
`buses`	NetworkFootprintBusList	Multicast address used by each bus
`exclusions`	NetworkFootprintAddressRangeList	Addresses that cannot be used
`collision`	NetworkFootprintCollision	Available addresses and conflict information

Named by NetworkFootprint.

`NetworkFootprintPort` STRUCTURE

Socket type, port selection mode and port value

Name	Type	Description
`kind`	NetworkFootprintPortKind	
`mode`	NetworkFootprintPortMode	
`value`	NetworkFootprintPortValue	

Named by NetworkFootprintPortList.

NetworkFootprintPortExclusions STRUCTURE

Port ranges that cannot be used

Name	Type	Description
`builtIn`	NetworkFootprintPortRangeList	
`configured`	NetworkFootprintPortRangeList	
`operatingSystem`	NetworkFootprintPortRangeList	

Named by NetworkFootprint.

NetworkFootprintPortRange STRUCTURE

Inclusive range from which a port may be selected

Name	Type	Description
`min`	u16	
`max`	u16	

Named by NetworkFootprintPortRangeList, NetworkFootprintPortValue.

NetworkFootprintSelfCollision STRUCTURE

Two buses allocated to the same multicast group

Name	Type	Description
`firstBus`	NetworkFootprintBusIdentity	
`secondBus`	NetworkFootprintBusIdentity	
`groupAddress`	string	

Named by NetworkFootprintSelfCollisionList.

OpFinished STRUCTURE

An operation is complete

Named by OpState.

OpNotFinished STRUCTURE

An operation is not complete

Name	Type	Description
`waitHint`	sen.Duration	How long to wait until the next try

Named by OpState.

OptionalTypeSpec STRUCTURE

data of an optional type

Name	Type	Description
`type`	string	qualified type name

Named by CustomTypeData.

`OptionalTypeSpecV4` STRUCTURE

Name	Type	Description
`type`	string	qualified type name

Named by CustomTypeDataV4.

`OptionalTypeSpecV5` STRUCTURE

Name	Type	Description
`type`	string	qualified type name
`hash`	u32	unique hash

Named by CustomTypeDataV5.

`PrecisionSleep` STRUCTURE

Be more precise at the expense of some CPU cycles.

Name	Type	Description
`veryCoarseGrainSleepTime`	sen.Duration	First set of sleeps. If 0 defaults to 7ms
`coarseGrainSleepTime`	sen.Duration	Second set of sleeps. If 0 defaults to 1ms

Named by SleepPolicy.

`ProcessData` STRUCTURE

Information about the process

Name	Type	Description
`processInfo`	ProcessInfo	general process information
`environment`	EnvironmentVarList	list of environment variables
`stacktrace`	StringList	the stack trace (if any)
`recentLogs`	StringList	the last logs emitted by the process

Named by ErrorReport.

`ProcessInfo` STRUCTURE

Basic information about a process

Name	Type	Description
`hostId`	u32	
`processId`	u32	
`sessionId`	u32	
`sessionName`	string	
`appName`	string	
`hostName`	string	
`osKind`	OsKind	
`osName`	string	
`cpuArch`	CpuArch	

Named by sen.components.ether.Hello, sen.components.ether.SessionPresenceBeam, ProcessData, Processes.

`PropertySpec` STRUCTURE

spec of a property

Name	Type	Description
`name`	string	name of the type
`description`	string	documentation
`category`	PropertyCategorySpec	category
`type`	string	qualified type name of the property
`transportMode`	TransportModeSpec	how to transport the data
`tags`	StringList	user-defined tags
`checkedSet`	bool	if setting the property is checked

Named by PropertySpecList.

`PropertySpecV4` STRUCTURE

Name	Type	Description
`memberHash`	u32	hash of the class member
`name`	string	name of the type
`description`	string	documentation
`category`	PropertyCategorySpec	category
`type`	string	qualified type name of the property
`transportMode`	TransportModeSpec	how to transport the data
`tags`	StringList	user-defined tags

Named by PropertySpecListV4.

`PropertySpecV5` STRUCTURE

Name	Type	Description
`memberHash`	u32	hash of the class member
`name`	string	name of the type
`description`	string	documentation
`category`	PropertyCategorySpec	category
`type`	string	qualified type name of the property
`transportMode`	TransportModeSpec	how to transport the data
`tags`	StringList	user-defined tags
`hash`	u32	unique hash

Named by PropertySpecListV5.

`QuantityTypeSpec` STRUCTURE

spec of a quantity type

Name	Type	Description
`elementType`	NumericType	type used to store the value
`unit`	UnitInfo	used unit
`minValue`	f64	discarded if >= maxValue
`maxValue`	f64	discarded if <= minValue

Named by CustomTypeData.

`QuantityTypeSpecV4` STRUCTURE

Name	Type	Description
`numericType`	NumericType	type used to store the value

Name	Type	Description
`unit`	UnitInfo	used unit
`minValue`	f64	discarded if >= maxValue
`maxValue`	f64	discarded if <= minValue

Named by CustomTypeDataV4.

`QuantityTypeSpecV5` STRUCTURE

Name	Type	Description
`numericType`	NumericType	type used to store the value
`unit`	UnitInfo	used unit
`minValue`	f64	discarded if >= maxValue
`maxValue`	f64	discarded if <= minValue
`hash`	u32	unique hash

Named by CustomTypeDataV5.

`QueueConfig` STRUCTURE

Name	Type	Description
`evictionPolicy`	QueueEvictionPolicy	what to do when the queue is full
`maxSize`	u64	0 means unbounded

Named by ComponentConfig.

`SenData` STRUCTURE

Information about Sen

Name	Type	Description
`version`	string	the Sen version
`compiler`	string	the compiler used to build Sen
`debugMode`	bool	true if it was build in debug mode
`buildTime`	string	the commit's timestamp, not the compiler's clock
`wordSize`	WordSize	architecture
`gitRef`	string	git ref
`gitHash`	string	git hash
`gitStatus`	GitStatus	status of the branch when build
`kernelParams`	KernelParams	Sen kernel parameters
`kernelProtocol`	u32	version of the kernel protocol
`transportProtocol`	u32	version of the transport protocol (if present)
`loadedComponents`	LoadedComponentList	components to load
`builtComponents`	BuiltComponentList	components to build

Named by ErrorReport.

`SequenceTypeSpec` STRUCTURE

spec of a sequence type

Name	Type	Description
`elementType`	string	qualified type name for stored values
`maxSize`	u64	discarded when 0
`fixedSize`	bool	

Named by CustomTypeData.

SequenceTypeSpecV4 STRUCTURE

Name	Type	Description
`elementType`	string	qualified type name for stored values
`maxSize`	u64	discarded when 0

Named by CustomTypeDataV4.

SequenceTypeSpecV5 STRUCTURE

Name	Type	Description
`elementType`	string	qualified type name for stored values
`maxSize`	u64	discarded when 0
`fixedSize`	bool	
`hash`	u32	unique hash

Named by CustomTypeDataV5.

SignalData STRUCTURE

Information about an unhandled signal

Name	Type	Description
`signalNumber`	i32	signal number
`signalName`	string	human-readable name

Named by ErrorData.

StructTypeFieldSpec STRUCTURE

a field of a struct

Name	Type	Description
`name`	string	field name
`description`	string	documentation
`type`	string	qualified type name held by this field

Named by StructTypeFieldSpecList.

StructTypeFieldSpecV4 STRUCTURE

Name	Type	Description
`name`	string	field name
`description`	string	documentation
`type`	string	qualified type name held by this field

Named by StructTypeFieldSpecListV4.

StructTypeFieldSpecV5 STRUCTURE

Name	Type	Description
`name`	string	field name
`description`	string	documentation
`type`	string	qualified type name held by this field

Name	Type	Description
`hash`	u32	unique hash

Named by StructTypeFieldSpecListV5.

`StructTypeSpec` STRUCTURE

spec of an struct type

Name	Type	Description
`fields`	StructTypeFieldSpecList	fields
`parent`	string	qualified type name the parent struct, empty means none

Named by CustomTypeData.

`StructTypeSpecV4` STRUCTURE

Name	Type	Description
`fields`	StructTypeFieldSpecListV4	fields
`parent`	string	qualified type name the parent struct, empty means none

Named by CustomTypeDataV4.

`StructTypeSpecV5` STRUCTURE

Name	Type	Description
`fields`	StructTypeFieldSpecListV5	fields
`parent`	string	qualified type name the parent struct, empty means none
`hash`	u32	unique hash

Named by CustomTypeDataV5.

`SystemSleep` STRUCTURE

Use the native system sleep

Named by SleepPolicy.

`UncaughtException` STRUCTURE

Information about an unhandled exception

Name	Type	Description
`exceptionKind`	ExceptionKind	kind of exception
`message`	string	message it provides (if any)

Named by ErrorData.

`UnitInfo` STRUCTURE

Describes a unit

Name	Type	Description
`name`	string	
`abbreviation`	string	
`category`	UnitCat	

Named by QuantityTypeSpec, QuantityTypeSpecV4, QuantityTypeSpecV5, UnitList.

`VariantTypeFieldSpec` STRUCTURE

a field of a variant

Name	Type	Description
`key`	u32	key/index of this type within the variant
`description`	string	documentation
`type`	string	qualified type name held by this field

Named by VariantTypeFieldSpecList.

`VariantTypeFieldSpecV4` STRUCTURE

Name	Type	Description
`key`	u32	key/index of this type within the variant
`description`	string	documentation
`type`	string	qualified type name held by this field

Named by VariantTypeFieldSpecListV4.

`VariantTypeFieldSpecV5` STRUCTURE

Name	Type	Description
`key`	u32	key/index of this type within the variant
`description`	string	documentation
`type`	string	qualified type name held by this field
`hash`	u32	unique hash

Named by VariantTypeFieldSpecListV5.

`VariantTypeSpec` STRUCTURE

spec of an struct type

Name	Type	Description
`fields`	VariantTypeFieldSpecList	

Named by CustomTypeData.

`VariantTypeSpecV4` STRUCTURE

Name	Type	Description
`fields`	VariantTypeFieldSpecListV4	

Named by CustomTypeDataV4.

`VariantTypeSpecV5` STRUCTURE

Name	Type	Description
`fields`	VariantTypeFieldSpecListV5	
`hash`	u32	

Named by CustomTypeDataV5.

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`BasicType` ENUMERATION

built-in others

Held as `u8`.

booleanType	0
stringType	1
durationType	2
timestampType	3

Named by BuiltInType.

`CompatibilityMode` ENUMERATION

What to do when a type published by another kernel does not match the local one

Held as `u8`.

relaxed	0
strict	1
disabled	2

Named by KernelParams.

`ComponentState` ENUMERATION

State of a kernel component

Held as `u8`.

unloaded	0
preloaded	1
loading	2
loaded	3
initializing	4
initialized	5
running	6
stopping	7
stopped	8
unloading	9
error	10

CpuArch ENUMERATION

Type of CPU

Held as u8.

x86	0
x64	1
arm2	2
arm3	3
arm4T	4
arm5	5
arm6T2	6
arm6	7
arm7	8
arm7a	9
arm7r	10
arm7s	11
arm64	12
mips	13
superH	14
ppc	15
ppc64	16
sparc	17
m68k	18
otherArch	19

Named by ProcessInfo.

ErrorCategory ENUMERATION

Category of a problem during component execution

Held as u8.

runtimeError	0
logicError	1
expectationsNotMet	2
ioError	3
other	4

Named by ExecError.

ExceptionKind ENUMERATION

Types of exceptions that can occur.

Held as u8.

runtime	0
logic	1
standard	2
unknown	3

Named by UncaughtException.

GitStatus ENUMERATION

Status of git

Held as u8.

clean	0
modified	1
unknown	2

Named by BuildInfo, SenData.

IntegralType ENUMERATION

built-in integrals

Held as u8.

uint8Type	0
int16Type	1
uint16Type	2
int32Type	3
uint32Type	4
int64Type	5
uint64Type	6

Named by EnumTypeSpec, EnumTypeSpecV4, EnumTypeSpecV5, NumericType.

`MethodConstnessSpec` ENUMERATION

method constness

Held as u8.

constant	0
nonConstant	1

Named by MethodSpec, MethodSpecV4, MethodSpecV5.

`NetworkFootprintBusSource` ENUMERATION

Bus source

Held as u8.

configured	0
supplied	1
runtime	2

Named by NetworkFootprintBus.

`NetworkFootprintPortKind` ENUMERATION

Type of socket opened by the process

Held as u8.

tcpAcceptor	0
udpUnicast	1
tcpSource	2

Named by NetworkFootprintPort.

`NetworkFootprintPortMode` ENUMERATION

Port allocation policy

Held as u8.

ephemeral	0
probe	1
pinned	2

Named by NetworkFootprintPort.

`OsKind` ENUMERATION

Type of operating system

Held as u8.

windowsOs	0
linuxOs	1
androidOs	2
appleOs	3
unixOs	4
posixOs	5
otherOs	6

Named by ProcessInfo.

`Priority` ENUMERATION

Thread execution priorities

Held as u32.

lowest	0
nominalMin	1
nominalMax	2
highest	3

Named by ComponentConfig.

`PropertyCategorySpec` ENUMERATION

how a property is seen by others

Held as u8.

staticRW	0
staticRO	1
dynamicRW	2
dynamicRO	3

Named by PropertySpec, PropertySpecV4, PropertySpecV5.

`PropertyRelationSpec` ENUMERATION

method property relation

Held as u8.

nonPropertyRelated	0
propertyGetter	1
propertySetter	2

Named by MethodSpec.

`QueueEvictionPolicy` ENUMERATION

Component queue eviction policy

Held as u32.

dropOldest	0
dropNewest	1

Named by QueueConfig.

`RealType` ENUMERATION

built-in reals

Held as u8.

float32Type	0
float64Type	1

Named by NumericType.

`RunMode` ENUMERATION

Run modes for the kernel

Held as u8.

realTime	0
virtualTime	1
virtualTimeRunning	2
startAndStop	3

Named by KernelParams.

`ThreadCreateErr` ENUMERATION

What can go wrong when creating a thread

Held as `u32`.

schedulerAlreadyStarted	0
invalidStackSize	1
invalidThreadFunction	2
invalidThreadFunctionArgument	3
invalidAffinity	4
tooManyThreads	5
internalOsError	6

`TransportModeSpec` ENUMERATION

how to transport information

Held as `u8`.

unicast	0
multicast	1
confirmed	2

Named by EventSpec, EventSpecV4, EventSpecV5, MethodSpec, MethodSpecV4, MethodSpecV5, PropertySpec, PropertySpecV4, PropertySpecV5.

`UnitCat` ENUMERATION

Type of measurement

Held as `u8`.

length	0
mass	1
time	2
angle	3
temperature	4
frequency	5
velocity	6
angularVelocity	7
acceleration	8
angularAcceleration	9
density	10
pressure	11
area	12
force	13
torque	14

Named by UnitInfo.

`WordSize` ENUMERATION

Native processor word size

Held as `u32`.

bits32	0
bits64	1

Named by BuildInfo, SenData.

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`BuiltInType` VARIANT

all built-in types

Type	Description
NumericType	
BasicType	

`CustomTypeData` VARIANT

all custom types

Type	Description
EnumTypeSpec	
QuantityTypeSpec	
SequenceTypeSpec	
StructTypeSpec	
VariantTypeSpec	
AliasTypeSpec	
OptionalTypeSpec	
ClassTypeSpec	

Named by CustomTypeSpec.

`CustomTypeDataV4` VARIANT

Type	Description
EnumTypeSpecV4	
QuantityTypeSpecV4	
SequenceTypeSpecV4	
StructTypeSpecV4	
VariantTypeSpecV4	
AliasTypeSpecV4	
OptionalTypeSpecV4	
ClassTypeSpecV4	

Named by CustomTypeSpecV4.

`CustomTypeDataV5` VARIANT

Type	Description
EnumTypeSpecV5	
QuantityTypeSpecV5	
SequenceTypeSpecV5	
StructTypeSpecV5	
VariantTypeSpecV5	
AliasTypeSpecV5	
OptionalTypeSpecV5	
ClassTypeSpecV5	

Named by CustomTypeSpecV5.

`NetworkFootprintPortValue` VARIANT

Known port number or range

Type	Description
u16	
NetworkFootprintPort Range	

Named by NetworkFootprintPort.

NumericType VARIANT

all built-in numeric types

Type	Description
IntegralType	
RealType	

Named by BuiltInType, QuantityTypeSpec, QuantityTypeSpecV4, QuantityTypeSpecV5.

OpState VARIANT

The status of an operation

Type	Description
OpFinished	done
OpNotFinished	not done

SleepPolicy VARIANT

Component sleep policy

Type	Description
PrecisionSleep	
SystemSleep	

Named by ComponentConfig, KernelParams.

Arrays

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ArgSpecList SEQUENCE

list of argument specs

```
sequence<ArgSpec> ArgSpecList;
```

Named by EventSpec, MethodSpec.

ArgSpecListV4 SEQUENCE

```
sequence<ArgSpecV4> ArgSpecListV4;
```

Named by EventSpecV4, MethodSpecV4.

ArgSpecListV5 SEQUENCE

```
sequence<ArgSpecV5> ArgSpecListV5;
```

Named by EventSpecV5, MethodSpecV5.

Buffer SEQUENCE

A general-purpose unbounded buffer

```
sequence<u8> Buffer;
```

Named by sen.components.jsonrpc.StaticFile.

BuiltComponentList SEQUENCE

```
sequence<BuiltComponentParams> BuiltComponentList;
```

Named by SenData.

CustomTypeSpecList SEQUENCE

list of custom types

```
sequence<CustomTypeSpec> CustomTypeSpecList;
```

Named by sen.components.jsonrpc.JsonRpcServer.

CustomTypeSpecListV4 SEQUENCE

```
sequence<CustomTypeSpecV4> CustomTypeSpecListV4;
```

CustomTypeSpecListV5 SEQUENCE

```
sequence<CustomTypeSpecV5> CustomTypeSpecListV5;
```

EnumeratorSpecList SEQUENCE

list of enumerators

```
sequence<EnumeratorSpec> EnumeratorSpecList;
```

Named by EnumTypeSpec.

EnumeratorSpecListV4 SEQUENCE

```
sequence<EnumeratorSpecV4> EnumeratorSpecListV4;
```

Named by EnumTypeSpecV4.

EnumeratorSpecListV5 SEQUENCE

```
sequence<EnumeratorSpecV5> EnumeratorSpecListV5;
```

Named by EnumTypeSpecV5.

EnvironmentVarList SEQUENCE

```
sequence<EnvVar> EnvironmentVarList;
```

Named by ProcessData.

EventSpecList SEQUENCE

list of event specs

```
sequence<EventSpec> EventSpecList;
```

Named by ClassTypeSpec.

EventSpecListV4 SEQUENCE

```
sequence<EventSpecV4> EventSpecListV4;
```

Named by ClassTypeSpecV4.

EventSpecListV5 SEQUENCE

```
sequence<EventSpecV5> EventSpecListV5;
```

Named by ClassTypeSpecV5.

LoadedComponentList SEQUENCE

```
sequence<LoadedComponentParams> LoadedComponentList;
```

Named by SenData.

MethodSpecList SEQUENCE

list of method specs

```
sequence<MethodSpec> MethodSpecList;
```

Named by ClassTypeSpec.

MethodSpecListV4 SEQUENCE

```
sequence<MethodSpecV4> MethodSpecListV4;
```

Named by ClassTypeSpecV4.

MethodSpecListV5 SEQUENCE

```
sequence<MethodSpecV5> MethodSpecListV5;
```

Named by ClassTypeSpecV5.

NetworkFootprintAddressRangeList SEQUENCE

```
sequence<NetworkFootprintAddressRange> NetworkFootprintAddressRangeList;
```

Named by NetworkFootprintMulticast.

NetworkFootprintBusList SEQUENCE

Buses included in a network footprint

```
sequence<NetworkFootprintBus> NetworkFootprintBusList;
```

Named by NetworkFootprintMulticast.

NetworkFootprintPortList SEQUENCE

Ports used by the process

```
sequence<NetworkFootprintPort> NetworkFootprintPortList;
```

Named by NetworkFootprint.

NetworkFootprintPortRangeList SEQUENCE

```
sequence<NetworkFootprintPortRange> NetworkFootprintPortRangeList;
```

Named by NetworkFootprintPortExclusions.

NetworkFootprintSelfCollisionList SEQUENCE

Multicast address conflicts found in the reported buses

```
sequence<NetworkFootprintSelfCollision> NetworkFootprintSelfCollisionList;
```

Named by NetworkFootprintCollision.

Processes SEQUENCE

Unbounded sequence of ProcessInfo

```
sequence<ProcessInfo> Processes;
```

PropertySpecList SEQUENCE

list of property specs

```
sequence<PropertySpec> PropertySpecList;
```

Named by ClassTypeSpec.

PropertySpecListV4 SEQUENCE

```
sequence<PropertySpecV4> PropertySpecListV4;
```

Named by ClassTypeSpecV4.

PropertySpecListV5 SEQUENCE

```
sequence<PropertySpecV5> PropertySpecListV5;
```

Named by ClassTypeSpecV5.

StringList SEQUENCE

Unbounded sequence of strings for general purpose

```
sequence<string> StringList;
```

Named by sen.components.jsonrpc.JsonRpcServer, sen.components.jsonrpc.NamedSelection, sen.components.jsonrpc.SessionInfo, BuiltComponentParams, ClassTypeSpec, ClassTypeSpecV4, ClassTypeSpecV5, ErrorData, KernelApi, ProcessData, PropertySpec, PropertySpecV4, PropertySpecV5.

StructTypeFieldSpecList SEQUENCE

list of struct fields

```
sequence<StructTypeFieldSpec> StructTypeFieldSpecList;
```

Named by StructTypeSpec.

StructTypeFieldSpecListV4 SEQUENCE

```
sequence<StructTypeFieldSpecV4> StructTypeFieldSpecListV4;
```

Named by StructTypeSpecV4.

StructTypeFieldSpecListV5 SEQUENCE

```
sequence<StructTypeFieldSpecV5> StructTypeFieldSpecListV5;
```

Named by StructTypeSpecV5.

U32List SEQUENCE

```
sequence<u32> U32List;
```

UnitList SEQUENCE

Unbounded sequence of unit information structs

```
sequence<UnitInfo> UnitList;
```

Named by KernelApi.

VariantTypeFieldSpecList SEQUENCE

list of variant fields

```
sequence<VariantTypeFieldSpec> VariantTypeFieldSpecList;
```

Named by VariantTypeSpec.

VariantTypeFieldSpecListV4 SEQUENCE

```
sequence<VariantTypeFieldSpecV4> VariantTypeFieldSpecListV4;
```

Named by VariantTypeSpecV4.

VariantTypeFieldSpecListV5 SEQUENCE

```
sequence<VariantTypeFieldSpecV5> VariantTypeFieldSpecListV5;
```

Named by VariantTypeSpecV5.

Optionals

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MaybeStacktrace		75
MaybeTransportProtocol		75
MaybeU64	optional value for a u64	76

MaybeException OPTIONAL

```
optional<UncaughtException> MaybeException;
```

MaybeF64 OPTIONAL

optional value for a f64

```
optional<f64> MaybeF64;
```

MaybeNetworkFootprintMulticast OPTIONAL

```
optional<NetworkFootprintMulticast> MaybeNetworkFootprintMulticast;
```

MaybeNetworkFootprintPortValue OPTIONAL

The value is not present when the port number is not known

```
optional<NetworkFootprintPortValue> MaybeNetworkFootprintPortValue;
```

MaybeSignal OPTIONAL

```
optional<SignalData> MaybeSignal;
```

MaybeStacktrace OPTIONAL

```
optional<StringList> MaybeStacktrace;
```

MaybeTransportProtocol OPTIONAL

```
optional<u32> MaybeTransportProtocol;
```

MaybeU64 `OPTIONAL`

optional value for a u64

```
optional<u64> MaybeU64;
```

log

16 types.

Fixed records

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Stderr	stderr sink	78
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`BasicFile` STRUCTURE

Basic file sink

Name	Type	Description
`fileName`	string	file to log to
`truncate`	bool	defaults to false
`createParentDir`	bool	true if we should create the parent dir first

Named by SinkType.

`ColorStderr` STRUCTURE

Colored stderr sink

Named by SinkType.

`ColorStdout` STRUCTURE

Colored stdout sink

Named by SinkType.

`Config` STRUCTURE

Overall component configuration

Name	Type	Description
`pattern`	string	default format (empty means default)
`level`	LogLevel	default global level for all
`sinks`	SinkConfigList	sinks to configure
`loggers`	LoggerConfigList	loggers to configure
`backtrace`	bool	log backtrace is printed on failure

Named by sen.kernel.KernelParams.

`LoggerConfig` STRUCTURE

Configuration used for a given spdlog logger.

Name	Type	Description
`name`	string	name of the logger
`sinks`	SinkNameList	name of the sinks to connect the logger to
`level`	LogLevel	default log level for this logger

Name	Type	Description
`pattern`	string	Formatting for the sinks in this logger. Does nothing if empty. Note that this replaces the format of the associated sinks, and this could affect other loggers using those sinks. The config is applied following the order of the Config.loggers field.

Named by LoggerConfigList.

`Null` STRUCTURE

Single-threaded null sink

Named by SinkType.

`RotatingFile` STRUCTURE

Rotating file sink

Name	Type	Description
`baseFileName`	string	
`maxSize`	u32	0 means use defaults
`maxFiles`	u32	0 means use defaults

Named by SinkType.

`SinkConfig` STRUCTURE

Configuration used for a given spdlog Sink.

Name	Type	Description
`name`	string	name of the sink
`singleThreaded`	bool	if true, single-threaded (defaults to false)
`pattern`	string	the pattern used for the formatting (empty means default).
`level`	LogLevel	default log level for this sink
`config`	SinkType	the type-specific configuration parameters

Named by SinkConfigList.

`Stderr` STRUCTURE

stderr sink

Named by SinkType.

`Stdout` STRUCTURE

stdout sink

Named by SinkType.

`Syslog` STRUCTURE

Syslog sink (only in linux)

Name	Type	Description
`identity`	string	
`option`	u32	
`enableFormatting`	bool	

Named by SinkType.

Enumerations

LogLevel	Severity/importance of a log message.	79

LogLevel ENUMERATION

Severity/importance of a log message.

Held as u8.

warn	0
err	1
critical	2
trace	3
debug	4
info	5
off	6

Named by sen.components.logmaster.LogMaster, sen.components.logmaster.Logger, Config, LoggerConfig, SinkConfig.

Variant records

SinkType	The type-specific configuration of the used spdlog Sink.	79

SinkType VARIANT

The type-specific configuration of the used spdlog Sink.

Type	Description
ColorStdout	
ColorStderr	
Stdout	
Stderr	
BasicFile	
RotatingFile	
Null	
Syslog	

Named by SinkConfig.

Arrays

LoggerConfigList	79
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LoggerConfigList SEQUENCE

```
sequence<LoggerConfig> LoggerConfigList;
```

Named by Config.

SinkConfigList SEQUENCE

```
sequence<SinkConfig> SinkConfigList;
```

Named by Config.

SinkNameList SEQUENCE

```
sequence<string> SinkNameList;
```

Named by LoggerConfig.

Built-in types

The types the language provides. Every one is written the same way on the wire wherever it appears.

Type	Meaning
`bool`	classic boolean
`f32`	32 bit floating point number
`f64`	64 bit floating point number
`i32`	32 bit signed integer
`sen.Duration`	a time duration
`sen.TimeStamp`	a point in time
`string`	unbounded string
`u16`	16 bit unsigned integer
`u32`	32 bit unsigned integer
`u64`	64 bit unsigned integer
`u8`	8 bit unsigned integer

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